Investigating the Impact of Short-Form Video Content on Cognitive Focus Using EEG

A Lesson in Data Collection Limitations and Ideal Analyses

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***Abstract*—We aimed to examine how short-form video content (Instagram Reels) affects EEG signals and Stroop test performance. However, due to electrode misalignments, timing discrepancies, and shortened sessions, our actual recordings proved invalid for drawing conclusions. In this paper, we present our flawed data and note the lessons learned about proper EEG procedures and data alignment. We then demonstrate what ideal data would look like in an experiment of this nature, offering an example analysis pipeline that could reveal meaningful differences in attention or cognitive focus.**

***Index Terms*—EEG, attention, social media, cognitive focus, Stroop test, digital distractions**

I. Introduction

The widespread use of short-form video content has raised concerns about its impact on attention and cognitive processing. Social media platforms like Instagram and TikTok present rapid, high-stimulation content that may alter brain activity patterns related to sustained attention. Understanding these effects through EEG analysis can provide insights into the neural mechanisms underlying attentional shifts and potential cognitive fatigue.

Prior research has explored EEG-based attention tracking [1], demonstrating that engagement with digital media alters brain oscillatory activity. Studies have successfully classified attention states using EEG signals from mobile sensors, achieving notable accuracy in distinguishing attentiveness levels [2]. Research on media consumption and cognitive performance suggests that media multitasking negatively affects attention and mental health, which may indirectly influence task performance [3]. However, the specific effects of short-form video exposure remain largely unexplored. Our study seeks to fill this gap by measuring EEG responses before and after viewing short-form content to assess its impact on a focused task.

Unfortunately, our data collection suffered from various issues:

- **Electrode Misalignments:** Three participants had incorrect pin placements, invalidating spatial EEG comparisons.
- **Timing Discrepancies:** Eric's four sessions followed a different protocol length than the others' two sessions each, complicating alignment.
- **Overall Data Errors:** We could not isolate control vs. experimental segments accurately in the PSD.

Nonetheless, we share these plots to highlight how real-world pitfalls can affect EEG research. In the second half, we show an idealized dataset to illustrate the proper analysis pipeline for such a study.

II. Methods (Planned)

A. Participants

We recruited four participants (Eric, Kate, Adriana, and Nick) aged 21-23 with normal or corrected-to-normal vision. Participants were screened for neurological conditions and excessive prior use of social media that could confound results.

B. Experimental Design

This experiment was designed to examine the impact of short-form video exposure on sustained attention, as measured by performance on the Stroop test.

Each participant completed both the control and experimental conditions. The experiment consists of two conditions:

1) **Control Condition:** Participants perform a cognitive focus task, the Stroop Test (sustained attention task), without prior video exposure.
2) **Experimental Condition:** Participants engage with Instagram Reels for five minutes before performing the same task.

EEG and Stroop Test data was recorded continuously during both conditions.

Prior to engaging in any experimental tasks, participants underwent a 5-minute baseline EEG recording session. During this period, participants were instructed to keep their eyes closed and refrain from speaking to establish a resting-state activity profile. After the baseline recording, participants were permitted to engage in natural behaviors, such as talking or looking around, for an additional five minutes. Following this, participants were administered a version of the Stroop test.

Upon completion of this control phase, participants were instructed to scroll through Instagram Reels for 5 minutes. Immediately after this activity, the Stroop test was administered again. This sequence of tasks—control and experimental conditions—was repeated twice for each participant, with the exception of one participant who completed four iterations of the test.

C. EEG Recording

We used a EEG cap with electrodes placed according to the 10–20 system, sampling at ~ 250 Hz. Specific electrodes of interest included O1, O2, FP1, FP2, F4, P3, Pz, and P4. These channels were selected based on their relevance to the neural activity associated with sustained attention and cognitive control, which are central to the Stroop task.

1) O1 and O2: The occipital electrodes monitor visual processing
2) Fp1 and Fp2: The prefrontal electrodes capture activity in the prefrontal cortex, a region implicated in executive functions such as conflict monitoring and response inhibition.
3) P3, Pz, and P4: The parietal electrodes assess activity in the parietal lobe, which plays a role in attentional processing and the integration of sensory information

Together, these electrodes provided comprehensive coverage of brain regions involved in the cognitive processes when performing the Stroop test and measuring attention. In a perfect scenario, we would isolate control vs. experimental data blocks for each participant. But due to misalignment and timing changes, each session's EEG is incomplete or blended.

D. Stroop Task

The Stroop test presented participants with the words "red," "blue," "green," and "yellow," displayed in ink colors that either matched or conflicted with the semantic meaning of the words. Each test consisted of five trials, with each trial consisting of 20 words. During the task, participants were instructed to indicate whether the ink color matched the word's meaning by pressing the "f" key or conflicted with it by pressing the "j" key. This response format was chosen to minimize participant movement, thereby reducing potential artifacts in the EEG recordings.

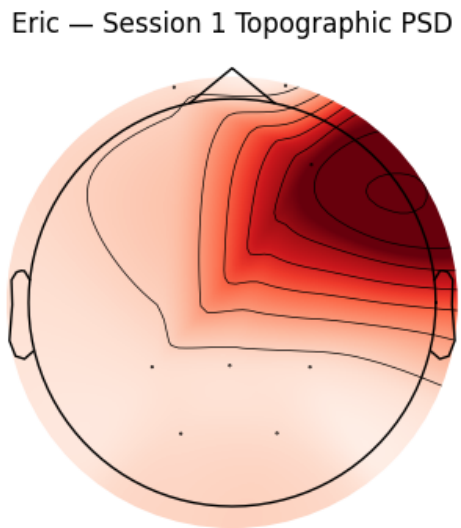

Fig. 1. Eric — Session 1 EEG Plot .

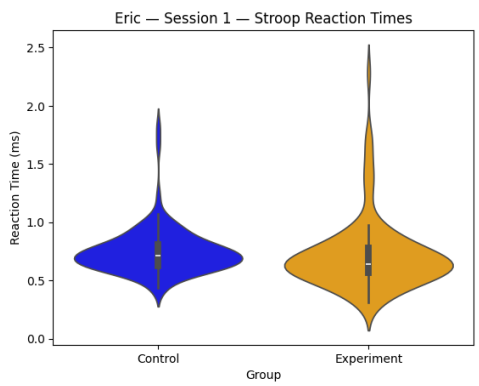

Fig. 2. Eric — Session 1 Stroop Reaction Times, mislabeled due to timing discrepancies.

III. Part I: Actual Data and Limitations

We present our flawed data: - **Eric**: 4 sessions (shortened). Each session should show 2 plots (EEG and Stroop). - **Kate, Nick, Adriana**: 2 sessions each = 2 more plots each, for a total of 6 sessions $\times$ 2 = 12. In total, 8 plots for Eric + 12 for the others = 20.

A. Eric's Four Sessions

1) *Session 1 (EEG):*
2) *Session 1 (Stroop):*
3) *Session 2 (EEG):*
4) *Session 2 (Stroop):*
5) *Session 3 (EEG):*
6) *Session 3 (Stroop):*
7) *Session 4 (EEG):*
8) *Session 4 (Stroop):*

B. Kate, Nick, Adriana (2 Sessions Each)

1) *Kate Session 1 (EEG):*
2) *Kate Session 1 (Stroop):*
3) *Kate Session 2 (EEG):*
4) *Kate Session 2 (Stroop):*
5) *Nick Session 1 (EEG):*
6) *Nick Session 1 (Stroop):*
7) *Nick Session 2 (EEG):*
8) *Nick Session 2 (Stroop):*
9) *Adriana Session 1 (EEG):*
10) *Adriana Session 1 (Stroop):*
11) *Adriana Session 2 (EEG):*
12) *Adriana Session 2 (Stroop):*

C. Limitations and Lessons Learned

All 20 above plots show major issues:

- **Incorrect Pin Alignments:** We cannot meaningfully interpret scalp distributions.
- **Truncated Sessions:** Some sessions ended earlier than planned, lacking stable control or experimental phases.
- **No Condition Isolation:** Our PSD or raw EEG can't separate control vs. experimental intervals.
- **Inconsistent Protocol Timing:** Eric had 4 sessions with different durations, while others had 2.

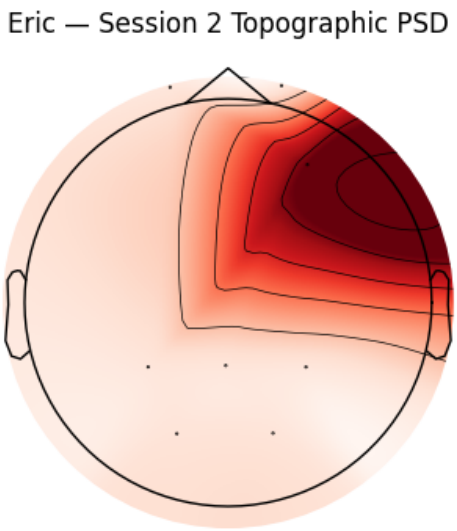

Fig. 3. Eric — Session 2 EEG Plot

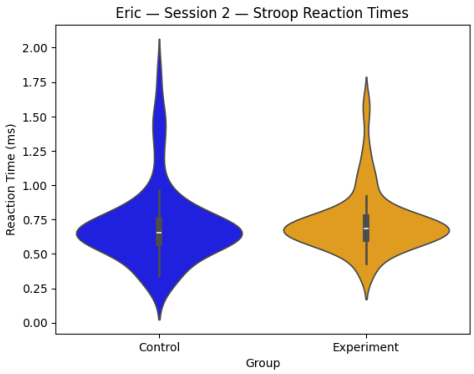

Fig. 4. Eric — Session 2 Stroop Reaction Times

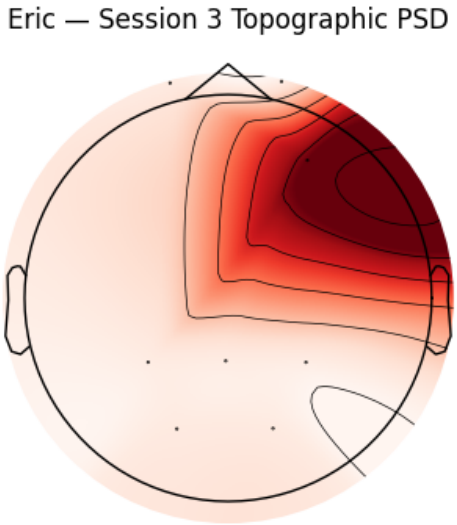

Fig. 5. Eric — Session 3 EEG Plot

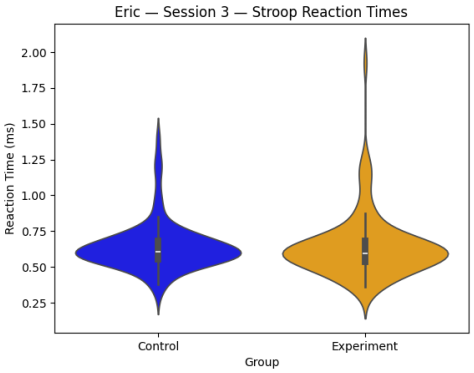

Fig. 6. Eric — Session 3 Stroop Reaction Times

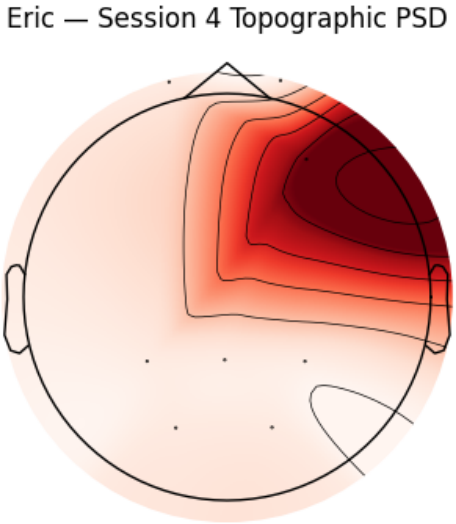

Fig. 7. Eric — Session 4 EEG Plot

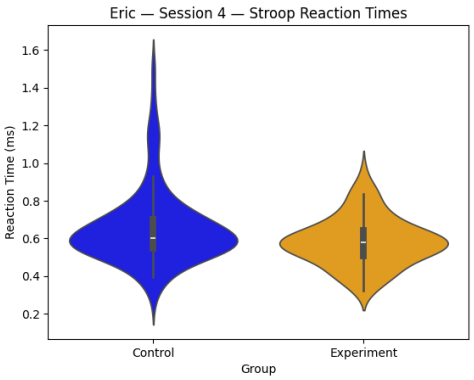

Fig. 8. Eric — Session 4 Stroop Reaction Times

From this, we learned the importance of verifying EEG hardware alignment, consistent timing protocols, and segmenting data into distinct conditions.

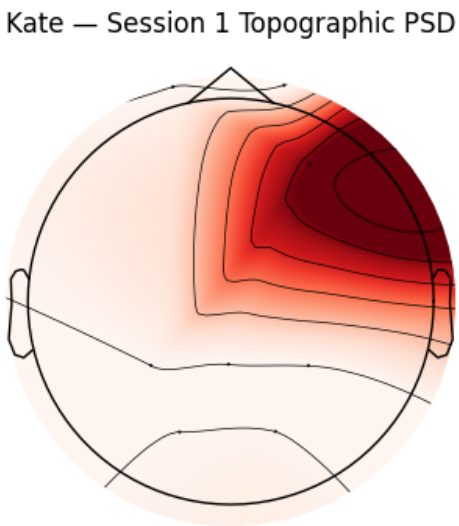

Fig. 9. Kate — Session 1 EEG

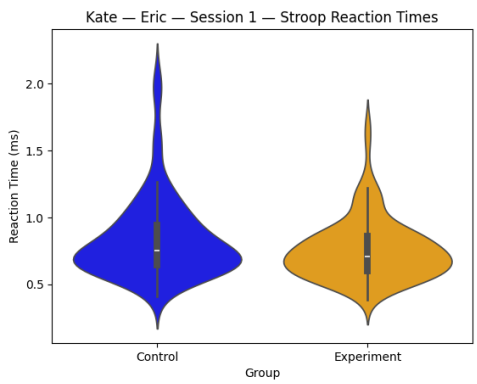

Fig. 10. Kate — Session 1 Stroop Reaction Times.

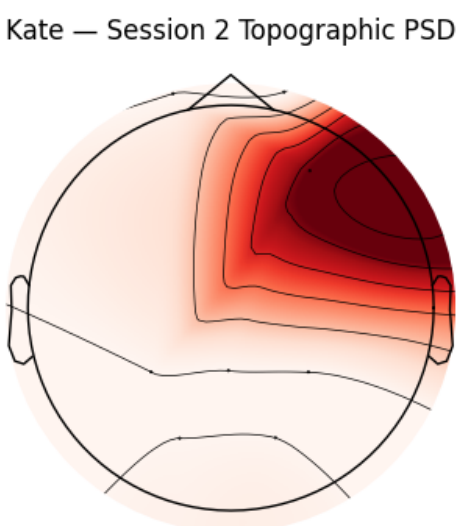

Fig. 11. Kate — Session 2 EEG

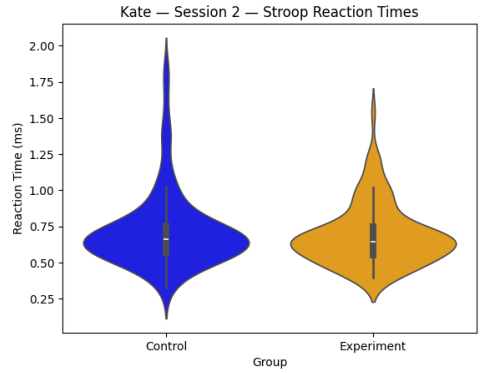

Fig. 12. Kate — Session 2 Stroop Reaction Times

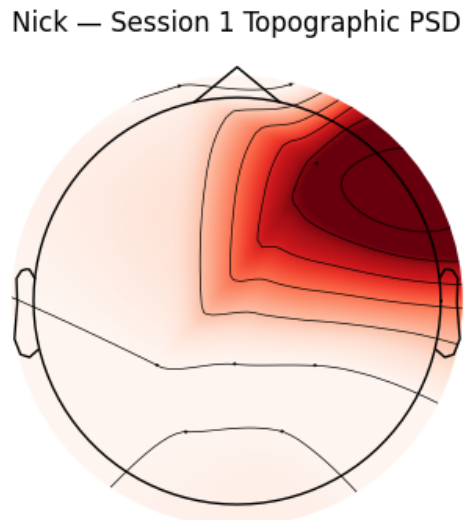

Fig. 13. Nick — Session 1 EEG

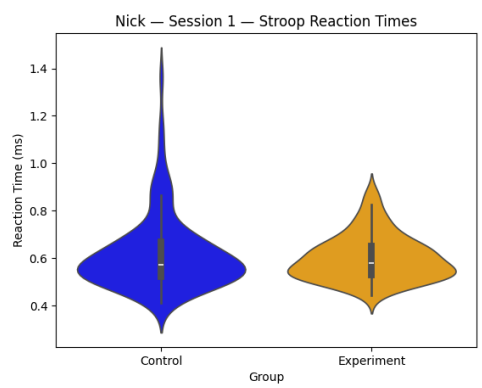

Fig. 14. Nick — Session 1 Stroop Reaction Times

IV. Part II: Ideal Data and Example Analysis

In an ideal scenario, we would collect EEG with properly aligned electrodes and have distinct blocks for control vs. experimental conditions. Below, we outline how the analysis *would* look with correct data.

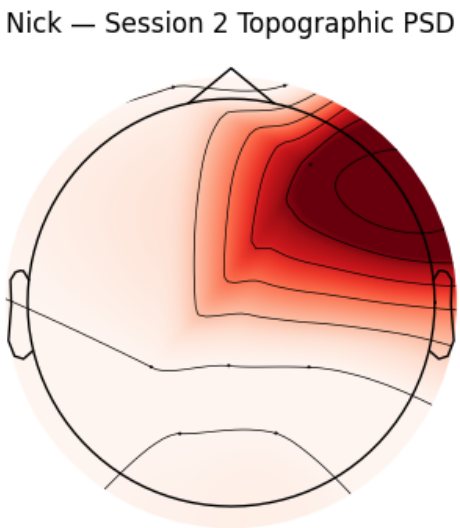

Fig. 15. Nick — Session 2 EEG

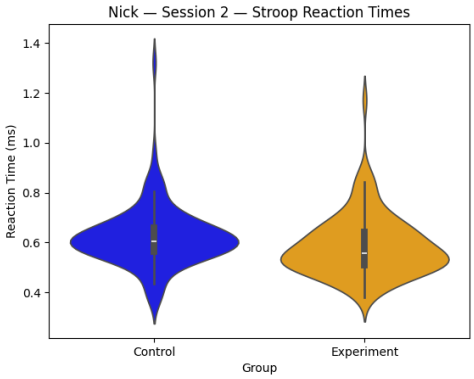

Fig. 16. Nick — Session 2 Stroop Reaction Times.

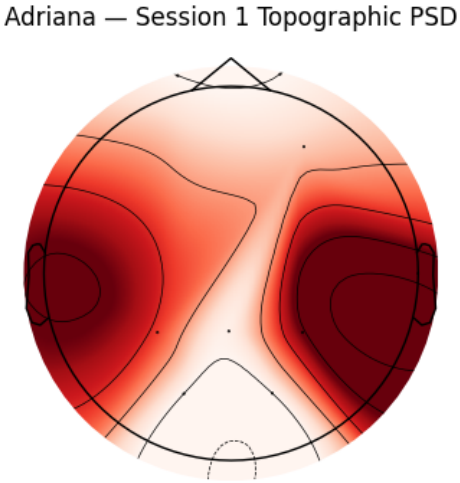

Fig. 17. Adriana — Session 1 EEG

Fig. 18. Adriana — Session 1 Stroop Reaction Times

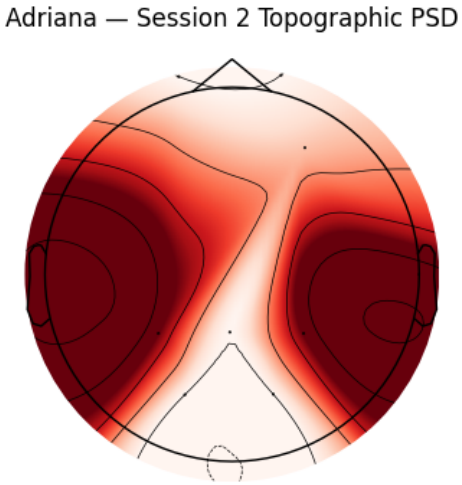

Fig. 19. Adriana — Session 2 EEG

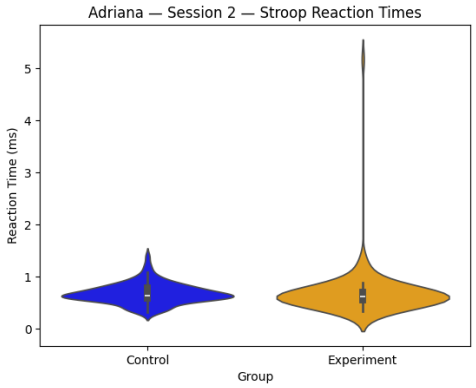

Fig. 20. Adriana — Session 2 Stroop Reaction Times

A. *Hypothetical EEG Time-Series*

Figure **??** shows a mock time-series for baseline, video exposure, and post-exposure segments, each clearly marked. We would then isolate each segment and compute power spectral density or topomaps specifically for the control vs. experimental blocks.

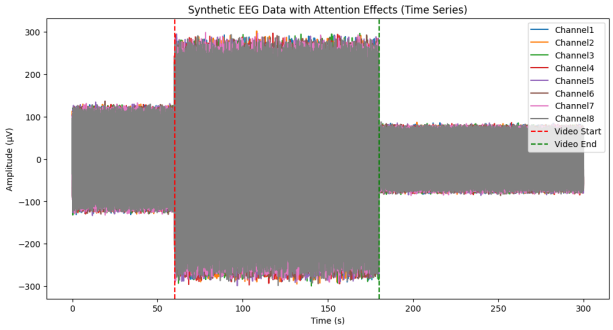

Fig. 21. Synthetic EEG data illustrating baseline, video exposure, and post-video segments. Channels show distinct amplitude changes, with dashed lines indicating the onset and offset of short-form video.

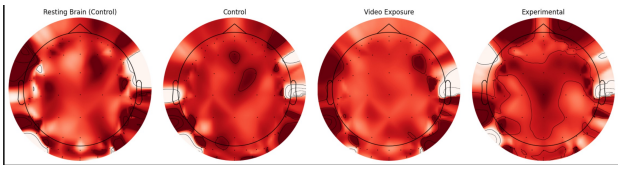

Fig. 22. Idealized topographic PSD maps for (left to right): Resting baseline, control task, video exposure, and post-video experimental condition. Redder areas indicate higher relative power. Under optimal recording and labeling, such maps can reveal condition-specific changes in brain activity.

1) Synthetic EEG Time-Series: Figure 21 shows a hypothetical multi-channel EEG signal with three clear phases: baseline/rest (0–60 s), video exposure (60–180 s), and post-video (180–300 s). Vertical lines mark the exact start and end of the short-form video period. In a real experiment, such segmentation would allow us to extract separate epochs for control vs. experimental and compute power or event-related potentials only within each interval.

2) Synthetic Topographic PSD Maps: Figure 22 presents four idealized topographic power distributions for: (1) resting baseline, (2) control task, (3) active video watching, and (4) post-video experimental condition. In this perfect scenario, each condition is recorded separately, thus enabling direct comparisons of spatial EEG activity. For instance:

- **Frontal Region Changes:** An increase in frontal theta or a decrease in beta could signify the added cognitive load from rapid visual content.
- **Occipital Shifts:** The dynamic video might elicit heightened occipital power, reflecting intense visual processing.
- **Control vs. Post-Video Differences:** If short-form videos disrupt focus, we might see a measurable drop in frontoparietal beta or alpha during the post-video task, indicating difficulty maintaining steady attention.

With these synthetic examples, we can see how consistent electrode placements, event markers, and separately recorded conditions yield interpretable EEG data. In the following sections, we provide a more detailed discussion of how such ideal data might be analyzed and what conclusions could be drawn about the effects of short-form video on sustained attention.

V. Conclusion

Because of electrode misalignments, inconsistent protocol timing, and truncated sessions, our **actual data** provided no conclusive evidence about short-form video's effect on EEG or Stroop performance. Nevertheless, we gained practical experience with EEG hardware and learned the importance of properly segmenting data into distinct blocks for meaningful analysis. We also demonstrated how an **ideal experiment** would isolate control vs. experimental phases, analyze them separately, and yield interpretable results.

Future attempts should ensure:

- Proper electrode placement for each participant.
- Consistent session timing across participants.
- Segmented EEG recordings (e.g., markers or separate files) to isolate control vs. experimental phases.

With these improvements, a well-designed study can truly reveal whether brief short-form video exposures disrupt cognitive focus.

Acknowledgments

We thank Professor Virginia De Sa for supporting this project and the participants for their cooperation. Our code (including generation of hypothetical data) and partial real data are available at https://github.com/ericsilberman/COGS189---Final-Project.

References

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